

THE ECONOMIC JOURNAL

JANUARY 2000

The Economic Journal, 110 (January), 1–26. © Royal Economic Society 2000. Published by Blackwell Publishers, 108 Cowley Road, Oxford OX4 1JF, UK and 350 Main Street, Malden, MA 02148, USA.

PORTFOLIO CHOICE IN THE PRESENCE OF BACKGROUND RISK*

John Heaton and Deborah Lucas

In this paper, we focus on how the presence of background risks – from sources such as labour and entrepreneurial income – influences portfolio allocations. This interaction is explored in a theoretical model that is calibrated using cross-sectional data from a variety of sources. The model is shown to be consistent with some but not all aspects of cross-sectional observations of portfolio holdings. The paper also provides a survey of the extensive theoretical and empirical literature on portfolio choice.

How do investors decide what fraction of their wealth to keep in stocks versus safe securities, and what explains cross-sectional variation in portfolio holdings? Despite the attention that the portfolio choice problem has received, precise answers have been elusive. What is clear empirically is the enormous variation across households in their portfolio allocations, ranging from the majority of households who hold no stocks, to a sizeable number who hold stocks almost exclusively. If this variation is to be explained in a utility maximising framework, it must be attributed to heterogeneity of preferences or of circumstances.

In this paper, we focus on how the presence of background risks – from sources such as labour income, proprietary income, and real estate – influence portfolio allocations. Using a variety of data sources, we characterise the joint distribution of these risk factors, and use this distribution to calibrate a theoretical model of portfolio choice. The theoretical model analysed here is a special case of Heaton and Lucas (1997), where we look at the effect of background labour-income risk on portfolio allocations. In that analysis, we found that the model was unable to generate significant cross-sectional variation in predicted portfolio composition as a function of the degree of background risk, even in the presence of transactions costs in the stock market. All agents held only stock almost all of the time. One reason for this is that labour income was assumed to have a low correlation with stock returns consistent with the observed relation between wage income and stock returns. Given this assumed labour income process, stock holdings tend to dominate bond holdings in the presence of a significant equity premium. In subsequent work (Heaton and Lucas, 1999), we observe that for those investors who hold a

* *The Economic Journal Lecture* at the Royal Economic Society Conference April 1998. We thank the National Science Foundation for financial support.

significant fraction of their wealth in stocks, proprietary business income is a large and more correlated background risk factor. We did not, however, formally model the effect of proprietary income risk on portfolio allocation. Here we demonstrate that assuming a background risk process that reflects proprietary business income and assuming somewhat higher risk aversion than in the earlier analysis is sufficient to generate considerable variation in predicted portfolio shares.

Another goal of the paper is to provide a survey of the extensive theoretical and empirical literature on portfolio choice. Sophisticated theoretical treatments of dynamic portfolio theory first appeared in the late 1960's, with papers by Mossin (1968), Merton (1969), and Samuelson (1969). These papers for the most part do not address the issue of background risk because, under the assumption of complete markets, background risk can be priced and capitalised into wealth. The statistical properties of background risk are then irrelevant to the allocation of wealth between risky stocks and risk-free bonds. Early empirical work by Friend and Blume (1975) suggested that these models were consistent with observed practice, in that calibrated versions of the portfolio share rules imply a fairly even division of wealth between stocks and bonds, assuming moderate risk aversion.¹ For these reasons, portfolio choice was thought to be a largely solved problem.

Recently there has been a resurgence of interest in the portfolio problem, both theoretically and empirically. There are several reasons for this trend. Growing interest in the implications of incomplete markets naturally leads to an examination of optimal portfolio behaviour in the presence of background risks, which then can affect allocation rules. Transactions costs and information problems, which have been shown to have significant effects on various types of economic decisions, also are likely to influence asset allocations. Finally, the equity premium puzzle suggests a reconsideration of the portfolio problem, since it suggests that portfolios should be tilted more heavily toward stocks than what we observe.² The fact that we restrict the analysis to decision-theoretic models rather than taking a general equilibrium approach is also due in part to the equity premium puzzle. Although one could look at portfolio choice in a general equilibrium framework, unless the model were capable of generating a realistic returns process the results on portfolio choice would also be suspect.

Empirically it has become apparent by looking more closely at cross-sectional data that few households follow portfolio policies that are consistent with simple theory. The list of difficult-to-explain phenomena includes the lack of participation in the stock market by the majority of United States households, the lack of diversification by many households who do hold stocks, and

¹ Interestingly, Friend and Blume argued that their findings supported the traditional theory because the share of stock in stocks, bonds, and liquid assets is fairly constant on average for sufficiently wealthy households. Similar data are used in more-recent studies to emphasise deviations from simple theory.

² The equity premium puzzle states that the high average spread between the returns on stocks and risk-free bills cannot be justified in an equilibrium version of the classical model used to explain portfolio choice.

most notably the propensity of wage earners to concentrate their stock holdings in the company in which they work. The cross-sectional data also suggest that most households face large and uninsured background risks in the form of labour income risk, proprietary income risk, and real estate risk. These observations have motivated a number of theoretical attempts (surveyed in Section 2) to reconcile these facts with optimising behaviour.

Beyond its descriptive value, the analysis of portfolio choice in the presence of background risk has potentially important implications for policy makers and practitioners. For instance in the debate over the privatisation of Social Security, a recurring question is whether unrestricted individuals are likely to make reasonable investment choices. What reasonable means, and whether there is evidence that households currently make good choices, can be better understood in the context of models that take heterogeneity seriously, and by closely scrutinising cross-sectional data. These analyses are also valuable to investment advisers, many of whom rely on the results as a basis for their recommendations.

The remainder of the paper is organised as follows. Section 1 presents empirical evidence on portfolio choice, and the nature of background risks that may influence portfolio decisions. Section 2 surveys the theoretical work on portfolio theory. Section 3 re-examines the question of whether calibrated theoretical models with background risks can explain some of the observed cross-sectional variations in portfolio holdings. Section 4 concludes.

1. Empirical Evidence

In aggregate, portfolio shares must reflect the aggregate supply of stocks, bonds and other financial assets. Hence, from an empirical perspective, portfolio theory is only useful to the extent that it can help explain the cross-sectional or temporal variations in portfolio composition. The most striking feature of the cross-sectional data is the heterogeneity in portfolio composition, as is documented in a number of papers including Bertaut (1994), Blume and Zeldes (1994), Friend and Blume (1975), Heaton and Lucas (1999), and Poterba (1993).

1.1. *Summary Statistics*

We begin by presenting a quantitative picture of heterogeneity that is consistent with the studies cited above. We compute summary statistics using the Survey of Consumer Finances (SCF). The SCF gives detailed financial and demographic information on a large panel of households, and provides population weights so that the results can be aggregated to represent the entire United States population.³ In reporting percentage portfolio allocations, the question arises as to what measure of wealth to use in the denomi-

³ For a more complete description of the data set and variable definitions, see Heaton and Lucas (1999).

nator. We define two measures. First, 'liquid assets' is the sum of stocks bonds, bills, and cash, held directly or indirectly, where indirect holdings consist of assets such as mutual funds and defined contribution pensions. 'Total financial wealth' consists of liquid assets, plus housing equity, other real estate, trusts, and the value of all private businesses. Although many studies use the narrower definition of liquid assets, we focus primarily on total financial wealth, since it is more meaningful when discussing the effect of background risks from these other wealth sources.

Table 1 presents summary statistics on the distribution of the share of stocks in financial wealth in 1989, 1992, and 1995, for all United States households. It shows that, consistent with the relative growth of the stock market, mean stock holdings, as a percentage of wealth, has increased over this period. It also illustrates the phenomenon of non-participation, which is discussed in more detail by Mankiw and Zeldes (1991) and Bertaut and Haliassos (1995). In all three years, stock holdings are zero in the 50th percentile, and only between 5.5% and 11% of wealth in the 75th percentile.

To try to understand the reasons for non-participation, it is natural to look for other characteristics that distinguish stockholders and non-stockholders. As a number of previous studies have shown, households with stock in their portfolio look considerably different than non-stock holders along several dimensions. We designate as 'Non Stockholders' those households with less than \$500 in stock and as 'Stockholders' those households with at least \$500 in stock. We set this cutoff above zero to avoid including households with incidental holdings, but the results are not very sensitive to this choice. Based on this classification, Table 2 reports the mean and median of total financial wealth, housing wealth, other real estate wealth, private business wealth, labour income, age and education in 1992, for each group. The results can be summarised by saying that stockholders are considerably wealthier in all asset categories, somewhat older, and better educated.

Due to the very limited wealth of non-participants, it is informative to also look at the cross-section of stock holdings, conditioning on those households who have more than \$500 in stock. In Table 3, we show that for this group, the average stock share in financial wealth is 0.23, and ranges from 0.06 at the 25th percentile to 0.33 at the 75th percentile. One possible explanation for this heterogeneity, which we explore in the next section, is that exposure to background risk varies systematically with stock holdings.

Table 1
Cross-sectional Variation of Share of Stocks in Financial Assets

Year	Percentile				Mean	Std. Dev.	Skewness
	25%	50%	75%	100%			
1989	0	0	0.055	1.0	0.132	0.300	2.33
1992	0	0	0.093	1.0	0.137	0.287	2.29
1995	0	0	0.113	1.0	0.151	0.299	2.05

Table 2
Characteristics of Stockholders and Non Stockholders (in 1992)

	Non stockholders		Stockholders	
	Mean	Median	Mean	Median
Total Financial Wealth \$	86,605	26,401	436,992	155,670
Housing Wealth \$	33,624	5,000	89,184	55,000
Other Real Estate Wealth \$	14,841	0	64,406	0
Private Business Wealth \$	16,272	0	98,704	0
Labour Income \$	17,502	3,500	42,359	33,600
Age	47.8	44	50.9	50
Education	12.4	12	14.7	16

1.2. Sources of Background Risk

As the theoretical models discussed in Section 2 below establish, uninsurable background risk can influence portfolio allocations, since it can change people's tolerance for stock market risk. The effects of background risks on stock holdings then depend on the correlation between the returns on stocks and the returns on the other assets. Tabulations using the SCF suggest that potentially important sources of background risk arise from labour income (which can be thought of as the income flow from human capital), owner-occupied and commercial real estate, and proprietary income from privately held businesses. Heaton and Lucas (1999) provide a detailed discussion of this issue, which we highlight in Table 4. Table 4 shows that the relative shares of these other assets in financial wealth vary across various age and net worth brackets. Net worth is defined as financial wealth net of all personal debt. The relative shares of the various assets also vary considerably within each category, and holdings are often highly skewed. Notice that housing is an important component of wealth for all age and wealth categories, and that private business wealth is an important component for the wealthy, particularly those under 65.

A growing literature examines these sources of background risk and their effect on portfolio composition. Labour income is perhaps the most obvious source of background risk that is large and difficult to insure or diversify. Several studies, using a variety of data sources, have focused on whether it

Table 3
*Cross-sectional Variation of Share of Stocks in Financial Assets,
 Stocks > \$500*

Year	Percentile			Mean	Std. Dev.	Skewness
	25%	50%	75%			
1992	0.06	0.16	0.33	0.23	0.26	5.17

Table 4
*Portfolio Shares Relative to Financial Assets, as a Function of Age and Net Worth,
 for Households with Stocks > \$500 (in 1992)*

Assett	Net Work (\$mn), Age					
	0.01–0.1		0.1–1		> 1	
	age < 65	age > 65	age < 65	age > 65	< 65	age > 65
Stocks	0.256/0.182	0.208/0.100	0.187/0.136	0.210/0.169	0.193/0.097	0.318/0.257
Bonds	0.055/0.002	0.071/0.011	0.053/0.007	0.080/0.013	0.055/0.010	0.121/0.020
Cash	0.147/0.080	0.206/0.082	0.106/0.049	0.233/0.188	0.078/0.044	0.115/0.060
Business	0.043/0.000	0.000/0.000	0.110/0.000	0.015/0.000	0.242/0.064	0.134/0.000
Housing	0.324/0.300	0.480/0.541	0.313/0.284	0.401/0.387	0.143/0.097	0.163/0.125
Other real estate	0.072/0.000	0.028/0.000	0.141/0.000	0.060/0.000	0.188/0.106	0.147/0.068

affects portfolio choice. The results have been mixed. Guiso *et al.* (1996) find that in Italian household data, background risk has a small affect on portfolio choice, while Hochguertel (1997) finds that in the Netherlands, households who report more risky income streams hold safer investment portfolios. Using United States data from the Panel Survey of Income Dynamics (PSID), Gakidis (1997) finds that households with a larger probability of realising extremely low wage income are less likely to participate in the stock market. For those who are participating, the probability of very low wages reduces the portion invested in stocks.

Motivated by the fact that private business ownership tends to be concentrated, somewhat illiquid, and returns are correlated with the stock market (see Table 5 below), several authors recently have focused on the effect of business ownership on portfolio choice. Heaton and Lucas (1999) find that households with more private business wealth hold fewer stocks relative to other liquid assets. Gentry and Hubbard (1998) examine the savings and investment decisions of entrepreneurs, and conclude that this group saves a larger fraction of wealth than other people, perhaps due to a precautionary demand for financing. Polkovnichenko (1998) finds that entrepreneurs appear to be less risk-averse than average, using a variable from the SCF that polls people about their risk tolerance.

From Table 4, one can see that housing is the most important asset class for many households. Like private businesses, housing is relatively illiquid and undiversified. Analysing the risk and return to housing is complicated by the unobservable consumption stream that it generates. Flavin and Yamashita (1998) consider housing as both an asset and a source of consumption services. They posit that the consumption demand for housing is likely to create a highly levered position in real estate for younger households. This levered position in a risky asset should affect their tolerance for stock market risk relative to older households who have paid down their mortgage. Flavin and Yamashita simulate what optimal portfolio shares should be in theory,

calibrating the model with estimates of the correlation between housing and other investments. However, they do not compare their predictions on portfolio holdings to the data. Goetzmann (1993) measures the risk and return to housing investments, and Brueckner (1997) discusses the interaction of consumption, investment, and home ownership.

Another source of concentrated risk is workers' investments in their employers' stock. To the extent that these holdings are voluntary, they cannot properly be considered uninsurable background risk. On the other hand, if they are the result of subsidised stock purchase programmes with restrictions on sales, defined contribution pension plans with concentrated holdings, or ESOPs, it is fair to classify them as a source of background risk. In any case, the quantities involved are significant. On average, 10.4% of common stock holdings reported in the SCF fall into this category. Heaton and Lucas (1999) report regression results indicating that holding other factors constant, the more of the employer's stock held, the lower is the share of other stocks in other liquid assets. Similarly, proprietary business holdings reduce the share of stock in liquid assets. Both these findings are consistent with the idea that the concentrated risk that is assumed by holding employer's stock discourages investment in other risky stocks.

1.3. *Quantifying Background Risk*

The results in the previous section suggest that there is considerable heterogeneity in exposure to background risk, and that households with greater exposure tend to hold a smaller share of stock in their portfolios. Here we provide some additional statistics to characterise the extent of heterogeneity in background risk. Our goal is to provide some guidance for the simulation exercise that we carry out in Section 3 and to assess better the extent to which background risk may be important. We focus on the differences across households in the volatility of the component of their income that can be considered background risk, along with the correlation of this income with stock returns.

We begin with aggregate data. Table 5 presents summary statistics for the properties of proprietary income, wage income and the returns to housing at the aggregate level in the United States. Wage income is measured as 'real wages and salaries', and proprietary income is real 'non-farm proprietary income', from the National Income and Product Accounts, 1947 to 1992. Housing statistics are from Goetzmann (1993). Proprietary income is taken as the best aggregate proxy for the income flow from private business holdings. Notice that aggregate proprietary income is both riskier and more correlated with stock returns than is aggregate wage income. Thus, there appears to be a common component to proprietary income that would induce individuals with greater proprietary income to hold proportionally less stock. The returns to housing investment at the aggregate level are also quite risky, but negatively correlated with the stock market.

Despite the aggregate tendency for income from various sources to be positively correlated, there is substantial heterogeneity in the correlation

Table 5

Properties of United States Quarterly Aggregate Wage Income, Proprietary Income and Returns to Housing

	Wage income*	Proprietary income*	Return to housing†
Standard deviation in annual units‡	8%	4%	4%
Correlation with stock returns§	−0.07	0.14	−0.20

* 'Wage income' is 'real wages and salaries' and 'proprietary income' is real 'non-farm proprietary income', both from 1947 to 1992 and obtained from the United States National Income and Product Accounts.

† Aggregate properties of returns to housing in the United States as reported by Goetzmann (1993). The data spans the period 1926 to 1989.

‡ Annualised standard deviation of (log) growth rates.

§ For wages and proprietary income: correlation of growth rate with CRSP value weighted return obtained from the CRSP data set. The value weighted return was converted to real units using the implicit price deflator for nondurables plus services. For return to housing: correlation with the S&P 500 return.

between individual income streams and aggregate returns that could account for some individual variation of share holdings. To assess the extent of this heterogeneity, we follow Heaton and Lucas (1999) and examine income from the 1979–90 Panel of Individual Tax Returns. The data set, referred to as the 'Tax Model', is an unbalanced panel with 45,000 randomly selected individuals having at least three years of observations. It contains most of the data reported in individual tax returns.⁴ The advantage of these data over data sets with a time series component like the PSID is that it has a substantial sample of individuals with significant amounts of financial wealth and income from other sources, such as proprietary businesses. (The SCF cannot be used for this analysis, since it does not have a significant time series dimension.)

From these data, we construct annual observations of individual wage and proprietary income. Proprietary income is constructed as adjusted gross income less wage income and income from stock and bond investments. This is essentially Schedule C and E income, which effectively includes commercial real estate investment as well. As in all panel data sets, there is substantial measurement 'noise' in these data. To minimise the effects of this noise we require that individuals have at least three years of data. Further, our sample of wage income for each household requires that each individual have at least \$5,000 of wage income in each year. To be included in the sample of individuals with proprietary income we required that an individual have at least \$500 of proprietary income in each year. This income data was first converted to real income using the Consumer Price Index, and then converted to (log) growth rates.

Based on this data, Fig. 1 presents histograms of the cross-section of standard deviations of wage and proprietary income growth, and the correlation of

⁴ These data were compiled by the University of Michigan and Ernst and Young. See Heaton and Lucas (1999) for a more complete description of these data.

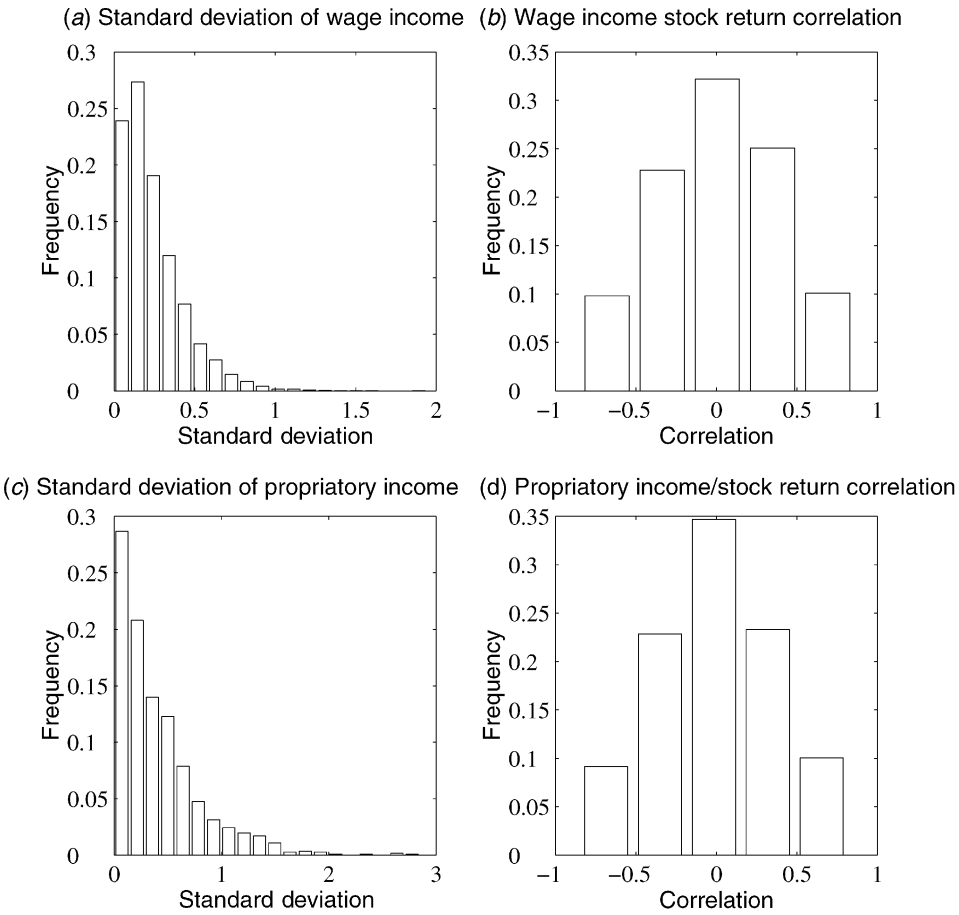


Fig. 1. *Cross-Sectional Properties of Wage and Proprietary Income*

these income growth rates with the return on the S&P 500. Each panel displays extensive heterogeneity in both the variability of individual income and the correlation of this income with stock returns. Across the two types of income, notice that the cross-sectional distribution of the standard deviation of proprietary income growth is more dispersed than that of wage income. For example, the median value for the standard deviation of wage income growth is 19% and for proprietary income growth it is 29%. There is a similar pattern to the cross-sectional distribution of the correlation with stock returns across the two sources of income.

Taken together, these results on aggregate and individual data imply substantial heterogeneity in the exposure to background risk. In Section 3 we explore whether a model calibrated to reflect the heterogeneity illustrated in Fig. 1 has the potential to generate the heterogeneity in portfolio choice observed in the data.

1.4. *Evidence on Holdings over the Life-cycle*

The evidence on stock holding over the life cycle is also mixed, with some authors emphasising the increase in stock holdings until retirement, and others the decline in stock holdings among the very old. Poterba and Samwick (1997) analyse the relationship between age and portfolio structure, with a focus on distinguishing between age effects and cohort effects. They find evidence of both age and cohort effects. Heaton and Lucas (1999) show that the age effect on stock holding is sensitive to whether wealth is broadly or narrowly defined. For instance, in Table 4 it appears that for wealthier households, older people hold a higher portion of financial assets in stocks. King and Leape (1987) also present evidence that households invest in an increasing number of asset classes as they age. They interpret this as a response to the cost of gathering information, which results in people learning about new assets only slowly over time.

2. Theory

Although our main interest is in models with background risk, a useful reference point is the literature on portfolio choice under complete markets.

2.1. *Complete Markets*

Merton (1969, 1971) provides the classic statement and solution of the dynamic portfolio optimisation problem.⁵ In these papers he derives conditions for optimal portfolio shares under a variety of assumptions about the returns process and utility specification. Explicit solutions are found when returns are generated by a Brownian motion process, and for HARA utility functions, a class that includes constant relative risk aversion (CRRA) and constant absolute risk aversion (CARA). Merton's papers contain a number of important results, including a two-fund separation theorem, and closed form solutions for the cases of constant relative risk aversion and constant absolute risk aversion.

The two-fund separation theorem states that given n assets with log-normally distributed prices, there exists a unique pair of 'mutual funds' consisting of a linear combination of the assets, such that independent of preferences, wealth distribution, or time horizon, investors will be indifferent between choosing from a linear combination of these two funds or a linear combination of the original n assets. This is useful because it simplifies the analysis of many assets to a two-asset case.

The closed form solution for constant relative risk aversion then treats the case of one risky and one risk-free asset. It is shown that the share of wealth invested in the risky asset, ω , is given by:

$$\omega = \frac{\alpha - r}{\sigma^2 \gamma} \quad (1)$$

⁵ Mossin (1968) and Samuelson (1969) also provide early treatments of this problem.

where r is the risk-free rate, α is the mean return on the risky asset, σ is the standard deviation of the risky asset, and γ is the coefficient of relative risk aversion. This holds for both the finite and infinite horizon cases. It is useful because it provides an example in which the share invested in the risky asset is affected neither by the level of wealth nor by the consumption decision. These properties make this utility function a convenient choice in many applications.

The closed form solution for the case of constant absolute risk aversion has the property that the total dollar value of wealth invested in the risky asset is constant regardless of wealth. This implies that as an investor becomes wealthier a decreasing fraction of his wealth is invested in the risky asset. This is generally considered an undesirable property of these preferences since richer people appear to invest more heavily in stocks and other risky assets than do those with low or moderate wealth. Nevertheless, this functional form is often chosen for convenience in continuous time models since it can be integrated against a normal distribution.

Merton's analysis of the portfolio problem relies upon dynamic programming techniques. An important development in this literature is the application of martingale techniques to the solution of the portfolio problem as developed by Cox and Huang (1989) and Karatzas *et al.* (1987). Martingale methods use the assumption of complete markets to convert the period-by-period budget constraint into a lifetime or atemporal budget constraint. Essentially a dynamic optimisation problem is rewritten as a static problem. This development, along with advances in dynamic programming, has enabled researchers to consider several important modifications of Merton's original problem.

First, the assumption of time-additive expected utility has been modified in a number of ways. Time nonseparability has been considered by adding habit persistence to preferences. The implications of this assumption have been considered, for example, by Constantinides (1990), Ryder and Heal (1973), and Sundaresan (1989). With habit formation individuals evaluate current consumption relative to an average of past consumption. This implies that risk aversion varies over time so that the optimal investment in the risky asset also varies. For example, Constantinides (1990) shows that Merton's result in (1) still holds except that γ is replaced by a time-varying coefficient of relative risk aversion.

It is well known that time variation in expected returns results in hedging demands for securities that are not reflected in (1), which was derived under the assumption of constant correlations. The form of this hedging demand is greatly affected by the intertemporal preferences of the agent. In this situation, a second important modification of preferences that has been considered is that of general 'recursive utility' which can accommodate both time and state-nonseparabilities in preferences. As shown by Epstein and Zin (1989), this preference specification allows for a distinction between risk aversion and the intertemporal elasticity of substitution. As discussed, for example, by Campbell and Viciera (1998), and Skiadas and Schroder (1998), this distinction is

important in assessing the effect of time-variation on returns on the hedging demands for assets.⁶

While these alternative utility specifications are conceptually useful, they do not provide directly testable empirical implications. This is because the models admit a variety of behaviours, and because preferences are not observable. Nevertheless, cross-sectional heterogeneity in preference parameters, in an environment with uninsurable background risk, does help to explain the large cross-sectional variation in portfolio allocations and possibly non-participation, as discussed in Section 3 below.

2.2. *Incomplete Markets*

A solution to the portfolio problem along the lines of Merton's applies quite generally in a complete markets setting, even if agents have other sources of risky income. This is because any future state and date contingent income can be priced and sold in the initial period, and added to total wealth. This total wealth is then invested in stocks and bonds according to the rules for portfolio optimisation. In contrast, when markets are incomplete some sources of future income cannot be capitalised. This greatly complicates finding analytic solutions to the portfolio problem.

Market incompleteness can be generated in a number of ways including limitations on the agent's ability to trade marketable securities such as short sales restrictions, transactions or information costs and non-marketable income streams. While our main concern below is on non-marketable income streams, a number of authors consider models in which all of the agent's wealth is represented by financial assets but there are trading limitations. For example, He and Pearson (1992), Karatzas *et al.* (1991), and Xu and Shreve (1992) examine the case where the payoffs to financial securities do not span all of the uncertainty in the economy, and there are short sales restrictions. The case of constraints on portfolio weights has been considered by Cvitanic and Karatzas (1992).

Transactions costs have been shown to have a significant impact on portfolio rules in models in which agents trade to smooth their consumption, but little impact when the purpose of trade is portfolio rebalancing. Constantinides (1986) and Davis and Norman (1990) consider the case in which markets are complete enough so that capital income is the only source of income risk. Constantinides finds that in this case transactions costs have a small effect on target portfolio allocations. This is because transactions costs discourage trading, so the costs are incurred only infrequently. In a model in which the motivation for trading is consumption smoothing rather than portfolio rebalancing, however, transactions costs can have a larger influence on portfolio choice because agents choose to trade frequently despite the costs. In Heaton

⁶ Other analyses of the portfolio problem under recursive utility include Fisher and Gilles (1998), Giovannini and Weil (1987), Hansen *et al.* (1998), and Svensson (1989).

and Lucas (1997) we find that with a differential transactions cost in the stock and bond market and uninsurable labour income risk, agents will tilt their portfolio toward the asset with lower trading costs.

An important form of market incompleteness is the presence of non-marketed assets, such as human capital, that generate a risky income stream that is not be spanned by traded assets. This type of background risk can potentially have an important effect on portfolio choice. Intuitively, one might expect that aversion to an additional risk, for instance to stock market risk, would be greater for agents with other sources of income risk. Whether this is predicted in theory, however, depends on the properties of the assumed utility function and the precise nature of the background risks. Pratt and Zeckhauser (1987) characterise utility functions with the intuitive property that background risk increases effective risk aversion. They call this property ‘proper risk aversion’. More precisely, proper risk aversion is defined as a condition on utility functions such that an undesirable lottery can never be made desirable by the presence of an independent, undesirable lottery. They show that this is a property of exponential, power, and logarithmic utility functions, so it holds for all of the most commonly used utility specifications.⁷

Although suggestive, these abstract analyses of background risk are, for the most part, of limited empirical relevance because of the strong statistical assumptions underlying the results. In practice, the background risks of interest are often correlated with portfolio risks, and have positive expected returns. For instance, the income risk from privately held businesses is likely to be positively correlated with the risk from common stocks, since both are influenced by business cycle conditions. Background risks also may not be stationary. Non-stationarity is a standard feature of lifecycle models, where labour income risk varies with age. Because most of these analytic characterisations rely on local approximations, it is also difficult to assess the magnitude of the effect of background risk in these models.

Dynamic theoretical models of portfolio choice in the presence of background risk and other constraints have been developed by a number of authors. For example, Cuoco (1997), Duffie *et al.* (1997), He and Pagès (1993), Svensson and Werner (1993), and Vila and Zariphopoulou (1997) present existence results and some characteristics of the solution to the continuous-time portfolio choice problem with stochastic labour income and borrowing constraints. The results of these papers provide important theoretical insights and suggest the potential importance of background risk for the portfolio choice problem. They are not, however, designed to provide a quantitative evaluation of the effects of background risk.

⁷ Gollier and Kimball (1997) further examine the relation between utility functions and background risk.

2.3. *Empirically-directed Incomplete Markets Theories*

To analyse portfolio choice in more complicated environments and to make predictions about magnitudes, it is often necessary to rely on numerical simulations and calibration. A number of more recent papers have taken this approach, including Bertaut and Haliassos (1997), Cocco *et al.* (1998), Gakidis (1997), Heaton and Lucas (1997), Koo (1995) and Viciara (1997).

In Heaton and Lucas (1997), we examine the quantitative implications of a decision-theoretic model of portfolio choice in which infinitely lived investors face uninsurable labour income risk. The sensitivity of savings and portfolio allocation rules to the utility function, the stochastic process for income and asset returns, and market frictions (transactions costs and short sale constraints) is examined. Koo (1995) considers a similar structure, but without transactions costs or habit formation. Further, Bertaut and Haliassos (1997), Cocco *et al.* (1998), Gakidis (1997) and Viciara (1997) consider a similar structure with a finite horizon.

The main results of these calibration exercises can be summarised as follows: Although one might expect that the presence of significant background risk from labour income would discourage stock holdings, this is not an implication of these models. In fact, all these studies find that under the assumption of CRRA time additive utility, investors borrow if they can, and invest all of their savings in stocks. This qualitative implication is robust to the introduction of significant transactions costs in the stock market. The reason these models imply such a strong preference for stocks is that any assumed floor on exogenous labour income functions as a risk-free asset, which substitutes for risk-free bond holdings. In other words, although labour income is risky, it limits bad outcomes relative to investment income, which significantly reduces effective risk aversion.

By using alternative preference specifications, it is possible to get some bond holdings in these calibrated models. Assuming habit formation, Heaton and Lucas (1997) find that it is possible to generate positive bond holdings in some states of the world due to the high effective risk aversion when the habit stock is high but income is low. An unfortunate side effect of habit formation, however, is that portfolio composition is volatile, varying significantly with the habit stock. Gakidas (1997) considers a finite horizon model with non-time separable preferences where the coefficient of relative risk aversion can be separated from the elasticity of intertemporal substitution. In this case, it is possible to get positive bond holdings by assuming a high elasticity of intertemporal substitution. The question then is how to realistically calibrate this elasticity parameter.

The above calibration papers focus on labour income as the primary source of background risk. In a similar model, Polkovnichenko (1998) considers an alternative source of background risk – income generated from privately held businesses, or ‘entrepreneurial income’. In part due to its higher correlation with stock market returns, he finds that the predicted demand for stocks is lower than in models that focus on labour income. In Section 3 below we

return to this type of calibration exercise to further explore how various types of background risk affect model predictions about portfolio allocations.

2.4. *Non-participation*

All the models discussed above imply that, if the mean rate of return on stocks is higher than that on bonds and if there are no fixed costs, all agents will hold at least a small amount of stocks. This is because with smooth preferences and return processes, utility is locally risk-neutral. This leaves unexplained the fact that, as discussed in Section 1 above, the majority of United States households hold no stocks. There have been several recent efforts to provide a theoretical explanation for the non-participation puzzle. In general, to get non-participation one must assume a fixed cost of stock market entry. This fixed cost may be tangible, for instance fixed brokerage fees or the cost of becoming sufficiently informed about the nature of risks and returns in the market, or unobservable, such as the psychic cost of putting savings at risk. Saito (1995) calibrates a model in which all wealth is held in the form of stocks or bonds, and calculates how large a fixed cost would be required at time 0 to result in some agents holding only bonds. He finds costs ranging from 3% to 54% of wealth, depending on the assumed risk aversion and size of equity premium. These numbers are high for the same reason that agents are predicted to put all of their wealth into stocks in the calibrated models discussed above – it is costly to forego the high equity premium.⁸ Polkovnichenko (1998) finds that in a model with heterogeneous risk aversion and heterogeneous idiosyncratic income risk, the fixed cost required to generate non-participation is potentially much lower.

2.5. *Life-cycle Effects*

Intuitively one might expect optimal portfolio weights to be a function of age, with older people holding relatively less stock, and this is certainly the recommendation of many financial advisors (Bodie and Crane, 1997). However, Samuelson (1969) showed that this intuition is contradicted under the normally assumed preference specifications, which imply no age variation in portfolio shares when capital income is a person's only source of income. Bodie *et al.* (1992) provide some theoretical underpinnings for the intuitive view. They show that if the opportunity to smooth income shocks by adjusting labour supply is greater for the young than for the old, in fact old people should hold less stock in their portfolios.⁹

⁸ Basak and Cuoco (1998) and Vissing-Jorgensen (1997) also explore these issues.

⁹ For further discussion of stock investment over the life cycle see, for example, Jagannathan and Kocherlakota (1996), and Cocco *et al.* (1998).

3. A Quantitative Model

In Section 2 we presented empirical evidence of the substantial cross-sectional heterogeneity in both portfolio composition and in background risk. We also provided empirical evidence linking these two observations. For example, we showed that households with significant amounts of risky proprietary income and higher mortgage balances tend to allocate less of their liquid assets to stocks. In order to assess more formally the potential quantitative importance of heterogeneity in background risk for portfolio choice, here we consider a dynamic model of portfolio choice with non-tradable income. The model is similar to Heaton and Lucas (1997), but here we use it to assess the implications of changing properties of background risk and the coefficient of relative risk aversion.¹⁰

3.1. Preferences and Markets

We assume an infinitely lived agent, with time additive utility. Preferences at time t are given by:

$$U_t = E_t \left[\sum_{\tau=0}^{\infty} \beta^{\tau} \frac{c_{t+\tau}^{1-\gamma} - 1}{1-\gamma} \right]; \gamma > 0. \quad (2)$$

At each time t , the agent receives non-tradable income, y_t . He chooses to invest s_{t+1} in stocks, b_{t+1} in bonds, and to consume c_t subject to the flow constraint:

$$c_t + s_{t+1} + b_{t+1} \leq s_t(1 + r_t^s) + b_t(1 + r^b) + y_t \quad (3)$$

where r_t^s is the return on stocks at time t , and r^b is the return on bonds. For simplicity, we assume that the return on bonds is constant over time.

In addition, we assume that the household faces a sequence of short-sale and borrowing constraints:

$$s_{t+1} \geq 0 \text{ and} \quad (4a)$$

$$b_{t+1} \geq 0. \quad (4b)$$

These restrictions imply that the agent cannot have negative financial wealth. This constraint could be relaxed without affecting the qualitative implications of the model.

To characterise a solution to the problem faced by the agent first notice that the optimal choices of consumption and security holdings must be such that inequality (3) holds as an equality. Hence at each time t , the agent chooses s_{t+1} and b_{t+1} , with c_t as a residual. These choices depend on two types of state variables. The first is given by current ‘flow’ wealth which is the sum of financial wealth, $w_t \equiv s_t(1 + r_t^s) + b_t(1 + r^b)$, plus non-tradable income, y_t .

¹⁰ The earlier analysis focused on labour income risk as the primary source of background risk. Unlike the earlier analysis, we do not consider transactions costs, nor do we consider habit persistence.

Let $Z_t \equiv w_t + y_t$ be this state variable, which is endogenous because it depends on the agent's past decisions. The second type of state variables are exogenous and describe the joint dynamics of non-tradable income and stock returns. As we describe in subsection 3.2 below we assume that the dynamics of these exogenous state variables are governed by a Markov chain and that the current state of this chain is given by the vector $\mathbf{X}_t$. The solution to the agent's optimisation problem is then given by two policy functions, $s(\mathbf{X}_t, Z_t)$ and $b(\mathbf{X}_t, Z_t)$ that give, respectively, the optimal stock and bond holdings as a function of the state variables.

The policy functions must be chosen such that the implied dynamics for consumption satisfy the first-order conditions for the problem. When the constraints (4) are not binding these are given by:

$$c_t^{-\gamma} = \beta E_t[c_{t+1}^{-\gamma} (1 + r_{t+1}^s)] \quad (5a)$$

and

$$c_t^{-\gamma} = \beta E_t[c_{t+1}^{-\gamma} (1 + r^b)], \quad (5b)$$

where the expectations at time t are a function of the state vector $\mathbf{X}_t$. If the agent is constrained by the short sale constraint, (5a) is replaced by (4a) with equality. Similarly if the agent is constrained by the borrowing constraint, (5b) is replaced by (4b) with equality.

It is not possible to calculate the optimal policy functions for this problem analytically. As discussed in Section 2, the theoretical problem has only been solved in closed form for special cases. Here the generality of the background income process and borrowing and short sales constraints preclude a simple solution. Instead, we use the numerical method described by Heaton and Lucas (1997) in which the true policy functions are approximated by piecewise linear functions of the state variables. As we discuss in subsection 3.2 below we assume that y_t grows over time. To accommodate this growth it is convenient to normalise the problem by dividing both the endogenous state variable Z_t and the policy functions by y_t . This normalisation induces stationarity in the endogenous state variable and is made possible by the assumed homotheticity of the agent's utility function.

3.2. Parameterisation of Processes for Income and Returns

The agent faces uncertainty in non-tradable income and from investing in stocks. To capture this uncertainty, let

$$\mathbf{X}_t \equiv \begin{bmatrix} \log(y_t/y_{t-1}) \\ \log(1 + r_t^s) \end{bmatrix}.$$

We assume that $\mathbf{X}_t$ follows a first-order autoregressive process:

$$\mathbf{X}_t = \boldsymbol{\mu} + \mathbf{A}\mathbf{X}_t + \mathbf{B}\boldsymbol{\epsilon}_t, \quad (6)$$

where $\boldsymbol{\epsilon}_t$ is a two-dimensional stochastic process that is independently and identically distributed over time as a normal random vector with mean zero and covariance matrix equal to the identity matrix. Although this is a simple

specification, it does capture some of the important features of income and helps to simplify computation of a solution to the model. Further it allows us to easily experiment with variations in the variability of non-tradable income, and the correlation between this income and stock returns.

To calibrate the process given by (6) we consider two specifications for the amount of non-tradable income risk. In the first, we follow Deaton (1991) and Koo (1995). We assume that the unconditional mean and standard deviation of the log growth rate of income are 1% and 15% respectively, where we interpret the time interval to be one year. We call this the 'low background risk' case. The standard deviation of income in this case is slightly smaller than the median standard deviation of the growth rate of wage income in our sample from the Tax Model. As Deaton (1992) points out, however, the higher median standard deviation in the cross section is likely due in part to measurement error. We saw previously that proprietary income is both riskier than wage income and is an important source of income for stockholders. To examine the effects of this riskier income we also consider a 'high background risk' case where we assume that the standard deviation of the growth rate in income is 25%. This assumed value is smaller than the median value of 29% we estimated from the Tax Model. Consistent with our assumption about wage income variability, we use a lower value than the median to reflect the possible extent of measurement error in the data. In both cases, we further assume that the log growth rate of non-tradable income is negatively correlated over time where the first-order autocorrelation is -0.4 . This assumption is consistent with the estimates presented by MaCurdy (1982) and, as discussed by Deaton (1992), this assumption induces the agent to follow a 'buffer-stock' savings pattern.

Stock returns are assumed to be stochastic, with a mean of 7.75% and standard deviation of 15.7%. This is consistent with historical data as discussed, for example, by Mehra and Prescott (1985). We assume that stock returns are not predictable over time. To examine the effect of heterogeneity in the correlation between income shocks and stock returns as illustrated by Fig. 1, we allow the innovations to non-tradable income to be correlated with stock returns. We consider a range of values for this implied correlation between -0.1 and 0.2 , which captures a significant range of the observed variation of the correlations estimated in our cross-sectional data. Each different combination of the variability of non-tradable income and its correlation with stock returns implies different values of μ , $\mathbf{A}$ and $\mathbf{B}$ of (6).

To reduce the dimension of the problem, for each assumption about the properties of income, we approximate the dynamics given by (6) with a four-state first-order Markov Chain. The parameters of these Markov Chain processes (values of the states and transitional probability matrices) are calculated using the techniques developed by Tauchen and Hussey (1991).

3.3. *Preference Parameters*

Throughout we assume that $\beta = 0.9$. A similar assumption is made by Deaton (1992) and Carroll (1992) in evaluating models of precautionary savings. The

subjective discount rate implied by this assumption is higher than the expected asset returns so that no savings would be accumulated in absence of non-tradable income. In the model with non-tradable income, the agent accumulates precautionary savings as a means to self-insure against shocks to non-tradable income.

Heaton and Lucas (1997) point out that when non-tradable income is relatively uncorrelated with stock returns, it acts like a bond in that it provides an income stream that mitigates the risk of a very low stock return. Unless the agent is sufficiently risk-averse, all savings will be carried out by holding stocks. Since the goal here is to examine the effects of variation in the properties of the background risk, we assume that γ is large enough so that there is at least some bond holdings even when non-tradable income is uncorrelated with stock returns. We consider two values for γ , 5 and 8. This also allows us to assess the effect of preference heterogeneity on stock and bond holdings.

3.4. Simulation Results

Summary statistics on the bond and stock holdings implied by numerical solutions to the model are presented in Table 6. In the table we present results for the two values for γ , for the two levels of risk in non-tradable income, and for the correlation between non-tradable income, ρ_{y,r_s} , equal to -0.1 , 0 , 0.1 and 0.2 . Recall that because of the growth in non-tradable income, each period we normalise bond and stock holdings by dividing them by current non-tradable income. The average level of these relative holdings are reported as 'Avg. bond holdings' for bonds and 'Avg. stock holdings' for stocks. We also report the average ratio of stock holdings to the total of stock plus bond

Table 6
Average Stock and Bond Holdings Predicted by the Model

	$\rho_{y,r_s} = -0.1$	$\rho_{y,r_s} = 0$	$\rho_{y,r_s} = 0.1$	$\rho_{y,r_s} = 0.2$
(a) $\gamma = 5$, Low background risk case				
Avg. bond holdings	0.03	0.05	0.07	0.11
Avg. stock holdings	1.12	1.14	1.15	1.15
Avg. proportion stock	0.98	0.97	0.96	0.93
(b) $\gamma = 8$, Low background risk case				
Avg. bond holdings	0.46	0.52	0.68	0.83
Avg. stock holdings	0.98	0.95	0.83	0.70
Avg. proportion stock	0.72	0.68	0.58	0.48
(c) $\gamma = 5$, High background risk case				
Avg. bond holdings	0.18	0.24	0.39	0.46
Avg. stock holdings	1.11	1.08	0.96	0.92
Avg. proportion stock	0.89	0.85	0.74	0.70
(d) $\gamma = 8$, High background risk case				
Avg. bond holdings	0.75	0.79	0.89	0.97
Avg. stock holdings	0.67	0.64	0.55	0.48
Avg. proportion stock	0.50	0.46	0.39	0.34

^a All securities holdings are normalised by current income to induce stationarity.

holdings as ‘Avg. proportion stock’. This gives the average fraction of financial assets that the agent holds in stocks.

Consider first the results in Panel (a) of Table 6 which are for $\gamma = 5$ and the low background risk case. When $\rho_{y,r_s} = 0$ the agent holds on average 97% of its savings in stocks. In contrast, consider the proportion of stock holding in Merton’s (1971) model where there is no non-tradable income (see (1)), under our assumptions about stock returns and when $\gamma = 5$:

$$\omega = \frac{0.0775 - 0.02}{0.157^2 \times 5} = 47\%.$$

The large average holding of stocks in our model occurs because the agent has a substantial flow of nonfinancial income that is uncorrelated with stocks.

The optimal allocation of stocks is quite sensitive to the assumed parameters of the model. When $\rho_{y,r_s} = 0$ and γ increases to 8 (Panel (b)), the share of stocks in the portfolio declines substantially to 68%. A similar decline in this share occurs for all the other values of ρ_{y,r_s} . The sensitivity of the results to the assumed value of γ is consistent with the significant empirical impact of risk attitude.

The stock allocation is also sensitive to the assumed properties of non-tradable income. For example when $\gamma = 5$ and $\rho_{y,r_s} = 0$, increasing the standard deviation of non-tradable income from 15% of the low background risk case, to 25% of the high background risk case reduces the share of stock from 97% to 85% (compare Panels (a) and (c) of Table 6). When $\gamma = 8$ the decline is even more dramatic going from 68% to 46% (compare Panels (b) and (d)).

For a fixed value of γ , an increase in the correlation between stock returns and non-tradable income reduces the attractiveness of stock returns as a means to hedge or self-insure against non-tradable income risk. For each specification of γ and assumed variability of non-tradable income, the third row of each panel in Table 6 illustrates the quantitative magnitude of this effect. For the smaller value of γ and low variability in the background risk, the effect of changes in the correlation is not large. When $\gamma = 8$, or in the high background risk case, the effect of the correlation is quantitatively more significant. Notice that under these same assumptions about γ and the variability of income, the model predicts significant bond holdings even though there is a significant equity premium. This result is consistent with the empirical observation that the typical household has significant holdings of relatively risk-free bonds. In this sense, the more extreme assumptions about preferences and variability of income may represent a plausible setting for assessing the impact of the correlation between non-tradable income and stock returns.

Analysing the results in this light, the model predicts that heterogeneity in the properties of background risk have a significant impact on stockholding. To assess the empirical relevance of this prediction, recall that at both the aggregate and individual level, proprietary income is much more variable than wage income and aggregate proprietary income is correlated with stock returns while wage income is not. Comparing the low background risk case with no

correlation to the high background risk case with $\rho_{y,r_s} = 0.2$, can be thought of as comparing a case where the individual has only wage income to a case where the individual has only proprietary income. When $\gamma = 8$ the share of stocks in the portfolio declines from 68% to 34%. This large effect could explain the significant impact of business value on the share of found in Heaton and Lucas (1999).

Despite these positive results, several aspects of the model are inconsistent with the empirical evidence. One of the most significant failings is illustrated in Fig. 2 which reports total savings and the portfolio rules (each relative to current non-tradable income) for different values of the sum of non-tradable

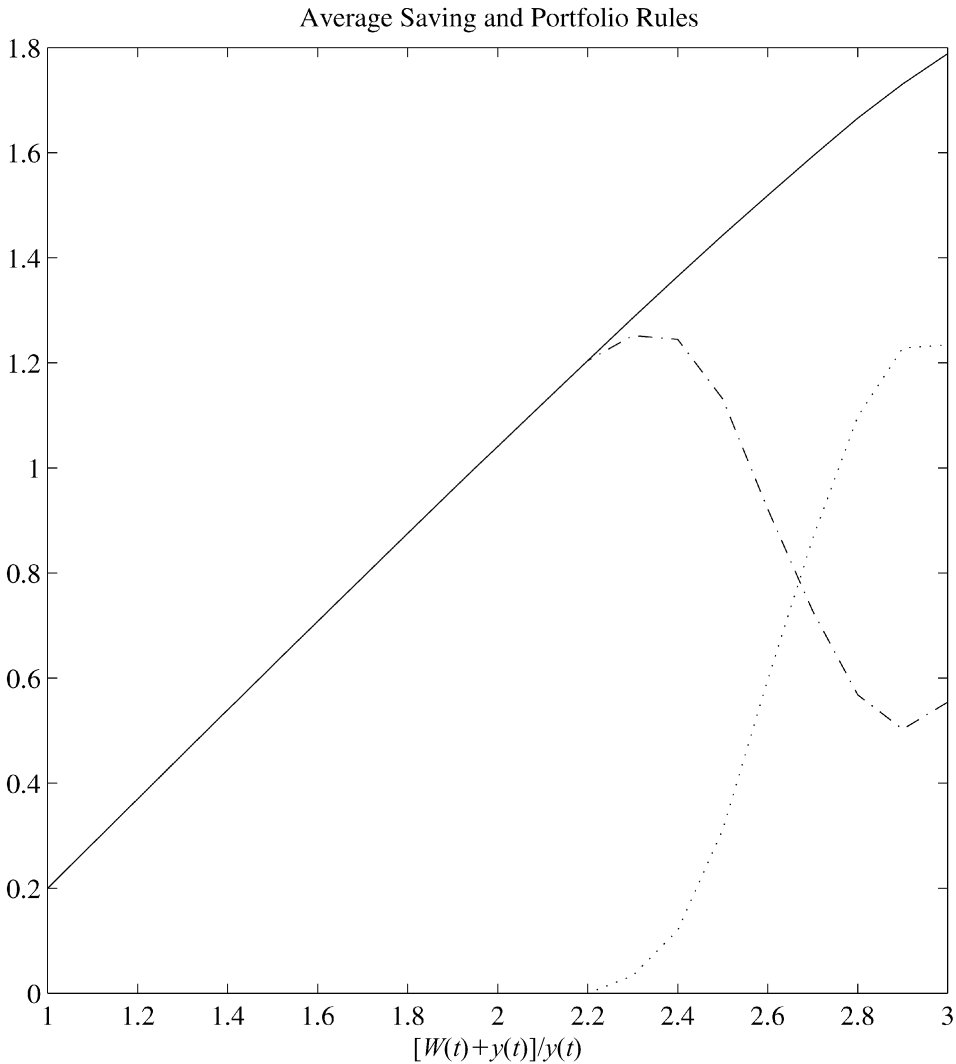


Fig. 2. *Portfolio Rules*, $\gamma = 8$, $\rho_{y,r_s} = 0$

income $y(t)$ plus financial wealth $w(t)$ relative to non-tradable income. Notice that for individuals where financial wealth is relatively unimportant, the ratio $[w(t) + y(t)]/y(t)$ is low, and all saving is done in the form of stock investments. Only when financial wealth becomes relatively more important does the individual begin to save using bonds. A more plausible result would be that individuals first build a buffer stock using bonds before shifting into stocks. Perhaps if the model were enriched to include a liquidity demand for short-term bonds and/or a fixed-cost of trading in the stock market, it would give a more reasonable prediction for stock holdings across all financial wealth levels.

Another prediction of the model that does not seem empirically reasonable is the magnitude and volatility of trading predicted by the model, as shown in Table 7. Savings are driven in this model by the agent's desire to self-insure against the significant background risk. The agent attempts to buffer this risk by trading in the bond and stock markets. One way to assess the plausibility of this mechanism is to examine the pattern of trading in the model. To this end, we report some statistics related to the amount of trading that occurs in the model and the variability of this trading over time. 'Avg. bond trades' and 'Avg. stock trades' are, respectively, the average amount of bond and stock trading implied by the model. Also 'Std. bond trades' and 'Std. stock trades' give the standard deviation of this trading over time. Finally, the extent of

Table 7
Other Predictions of the Model

	$\rho_{y,r_s} = -0.1$	$\rho_{y,r_s} = 0$	$\rho_{y,r_s} = 0.1$	$\rho_{y,r_s} = 0.2$
(a) $\gamma = 5$, Low background risk case				
Avg. bond trades	0.03	0.05	0.07	0.11
Avg. stock trades	0.12	0.13	0.14	0.18
Std. bond trades	0.08	0.11	0.14	0.20
Std. stock trades	0.16	0.18	0.21	0.25
Std. cons. growth	0.10	0.10	0.10	0.11
(b) $\gamma = 8$, Low background risk case				
Avg. bond trades	0.27	0.27	0.27	0.24
Avg. stock trades	0.33	0.32	0.31	0.28
Std. bond trades	0.31	0.29	0.23	0.18
Std. stock trades	0.35	0.33	0.27	0.21
Std. cons. growth	0.09	0.09	0.09	0.09
(c) $\gamma = 5$, High background risk case				
Avg. bond trades	0.16	0.19	0.29	0.29
Avg. stock trades	0.29	0.31	0.38	0.37
Std. bond trades	0.19	0.21	0.23	0.21
Std. stock trades	0.28	0.27	0.28	0.25
Std. cons. growth	0.15	0.15	0.15	0.15
(d) $\gamma = 8$, High background risk case				
Avg. bond trades	0.24	0.21	0.18	0.17
Avg. stock trades	0.31	0.27	0.22	0.19
Std. bond trades	0.16	0.14	0.13	0.14
Std. stock trades	0.20	0.18	0.14	0.12
Std. cons. growth	0.13	0.13	0.13	0.13

All securities holdings are normalised by current income to induce stationarity.

consumption smoothing achieved by the agent is reflected in the statistic ‘Std. cons. growth’ which is the standard deviation of consumption growth. In all of the simulations reported in Table 7 a significant fraction of both the bond and stock portfolio turns over every year on average. For example, when $\gamma = 8$ and $\rho_{y, r_s} = 0$, a third of the stock portfolio is traded each year. Further, the standard deviation of this trading is of the same order of magnitude as the trading itself. As reported by Heaton and Lucas (1997), adding trading costs to the model does reduce this trading somewhat, but the baseline amount of trading is quite extreme.

4. Conclusion

In this paper, we further explore the effects of background risk on portfolio choice in a calibrated decision-theoretic model. We find that realistic differences in background risk, due to cross-sectional variation in factors such as labour income, proprietary business risk, and in ownership of employer’s stock, can generate significant differences in predicted holdings. This is consistent with our earlier empirical findings (Heaton and Lucas, 1999) that households with significant business holdings or ownership of their employer’s stock tend to hold a smaller fraction of their liquid financial assets in the form of stocks. These new calibration results, combined with existing results in the literature surveyed in the paper, point to an important role for background risk in understanding portfolio decisions taken by households.

Although we find that background risk helps to explain portfolio decisions, several important issues have been left unresolved. The model is not able to explain the overall low level of participation in the stock market. It also cannot explain the low level of stock investment by those with modest financial wealth but significant human capital who do participate. Future theoretical research is needed to enrich the model to capture these features of household level data.

More generally, a better understanding of the implications of background risk for portfolio decisions is important for a number of issues. First, changes in the general economic environment due to financial innovation or revisions to government programmes, for example, will result in differences in household exposure to background risk over time. These changes will potentially lead to changes in how savings are allocated towards risky versus risk-free investment. Although we did not consider this issue in this paper, these same changes in the economic environment will also lead to changes in the level of savings as emphasised by the literature on precautionary saving. Second, there are potentially important implications for corporate finance to be drawn from the results of this paper. For example, the financial structure chosen by an entrepreneur will affect his exposure to background risk. There should therefore be an interesting relationship between the financial structure of small businesses and the portfolio decisions of the owners of those businesses. Third, the implications of this paper are important for asset pricing since background

risks that have an effect on the demand for risky assets will have an effect on the prices of those assets.

University of Chicago and NBER

Northwestern University and NBER

Date of receipt of first submission: January 1999

Date of receipt of final submission: July 1999

References

- Basak, S. and Cuoco, D. (1998). 'An equilibrium model with restricted stock market participation.' *Review of Financial Studies*, vol. 11, pp. 309–41.
- Bertaut, C. (1994). 'Who holds stocks in the U.S.? An empirical investigation.' manuscript, Board of Governors of the Federal Reserve System.
- Bertaut, C. and Haliassos, M. (1995). 'Why do so few hold stocks?' *ECONOMIC JOURNAL*, vol. 105, pp. 1110–29.
- Bertaut, C. and Haliassos, M. (1997). 'Precautionary portfolio behavior from a life-cycle perspective.' *Journal of Economic Dynamics and Control*, vol. 21, pp. 1511–42.
- Blume, M. and Zeldes, S. (1993). 'The structure of stockownership in the U.S.' manuscript, Wharton.
- Bodie, Z. and Crane, D. (1997). 'Personal investing: advice, theory, and evidence.' *Financial Analysts Journal*, vol. 53, pp. 13–33.
- Bodie, Z., Merton, R. and Samuelson, W. (1992). 'Labor supply flexibility and portfolio choice in a life-cycle model.' NBER Working Paper No. 6.
- Brueckner, J. (1997). 'Consumption and investment motives and the portfolio choices of homeowners.' *Journal of Real Estate Finance and Economics*, vol. 15, pp. 159–80.
- Campbell, J. and Viciera, L. (1998). 'Consumption and portfolio decisions when expected returns are time varying.' working paper, Department of Economics, Harvard University.
- Carroll, C. (1992). 'The buffer-stock theory of saving: some macroeconomic evidence.' *Brookings Papers on Economic Activity*, vol. 2, pp. 61–156.
- Cocco, J., Gomes, F. and Maenhout, P. (1998). 'Consumption and portfolio choice over the life-cycle.' manuscript, Harvard University.
- Constantinides, G. (1986). 'Capital market equilibrium with transactions costs.' *Journal of Political Economy*, vol. 94, pp. 842–62.
- Constantinides, G. (1990). 'Habit formation: a resolution of the equity premium puzzle.' *Journal of Political Economy*, vol. 98, pp. 519–43.
- Cox, J. and Huang, C.-F. (1989). 'Optimal consumption and portfolio policies when asset prices follow a diffusion process.' *Journal of Economic Theory*, vol. 49, pp. 33–83.
- Cogley, T. (1998). 'Idiosyncratic risk and the equity premium: evidence from the consumer expenditure survey.' manuscript.
- Cuoco, D. (1997). 'Optimal consumption and equilibrium prices with portfolio cone constraints and stochastic labor income.' *Journal of Economic Theory*, vol. 72, pp. 33–73.
- Cvitanic, J. and Karatzas, I. (1992). 'Convex duality in constrained portfolio optimization.' *Annals of Applied Probability*, vol. 2, pp. 767–818.
- Davis, M. and Norman, A. (1990). 'Portfolio selection with transactions costs.' *Mathematics of Operations Research*, vol. 15, pp. 676–713.
- Deaton, A. (1991). 'Savings and liquidity constraints.' *Econometrica*, vol. 59, pp. 1221–48.
- Duffie, D., Fleming, W., Soner, H. and Zariphopoulou, T. (1997). 'Hedging in incomplete markets with HARA utility.' *Journal of Economic Dynamics and Control*, vol. 21, pp. 753–82.
- Epstein, L. and Zin, S. (1989). 'Substitution, risk aversion and the temporal behavior of consumption and asset returns: a theoretical framework.' *Econometrica*, vol. 57, pp. 937–69.
- Fisher, M. and Gilles, C. (1998). 'Consumption and asset prices with recursive preferences.' working paper.
- Flavin, M. and Yamashita, T. (1998). 'Owner-occupied housing and the composition of the household portfolio over the life cycle.' NBER Working Paper no. 6389.
- Friend, I. and Blume, M. (1975). 'The demand for risky assets.' *American Economic Review*, vol. 65, pp. 900–22.
- Gakidis, H. (1997). 'Stocks for the old? Earnings uncertainty and life-cycle portfolio choice.' manuscript, MIT.

- Gentry, W. and Hubbard, R. G. (1998). 'Why do the wealthy save so much?' manuscript, Columbia University.
- Giovannini, A. and Weil, P. (1989). 'Risk aversion and intertemporal substitution in the capital asset pricing model.' NBER working paper no. 2824, Cambridge, MA.
- Goetzmann, W. (1993). 'The single family home in the investment portfolio.' *Journal of Real Estate Finance and Economics*, vol. 6, pp. 201–222.
- Gollier, C. and Kimball, M. (1997). 'Toward a systematic approach to the economic effects of uncertainty I: comparing risks.' Manuscript, IDEI, Toulouse, France.
- Gollier, C. and Zeckhauser, R. J. (1996). 'Horizon length and portfolio risk.' NBER Technical Working Paper 216.
- Guiso, L., Jappelli, T. and Terlizzese, D. (1996). 'Income risk, borrowing constraints and portfolio choice.' *American Economic Review* vol. 86, pp. 158–72.
- Hansen, L. P., Sargent, T. and Tallarini, T. (1998). 'Robust permanent income and pricing.' manuscript, Stanford University.
- He, H. and Pagès, H. (1993). 'Labor income, borrowing constraints, and equilibrium asset prices: a duality approach.' *Economic Theory*, vol. 3, pp. 663–96.
- He, H. and Pearson, N. (1992). 'Optimal consumption and portfolio policies with an infinite horizon: the infinite dimensional case.' *Journal of Economic Theory*, vol. 55, pp. 259–304.
- Heaton, J. and Lucas, D. (1997). 'Market frictions, savings behavior, and portfolio choice.' *Macroeconomic Dynamics*, vol. 1, pp. 76–101.
- Heaton, J. and Lucas, D. (1999). 'Asset pricing and portfolio choice: the importance of entrepreneurial risk.' *Journal of Finance*, forthcoming.
- Hochguertel, S. (1997). 'Precautionary motives and portfolio decisions.' manuscript, Tilburg University.
- Hubbard, R. G., Skinner, J. and Zeldes, S. (1995). 'Precautionary saving and social insurance.' *Journal of Political Economy*, vol. 103, pp. 360–99.
- Jagannathan, R. and Kocherlakota, N. (1996). 'Why should older people invest less in stocks than younger people?' *Quarterly Review*, Minneapolis Federal Reserve Bank, Summer, pp. 11–23.
- Karatzas, I., Lehotczky, J., Shreve, S. and Xu, G.-L. (1991). 'Martingale and duality methods for utility maximization in an incomplete market.' *SIAM Journal of Control and Optimization*, vol. 29, pp. 702–730.
- Karatzas, I., Lehotczky, J. and Shreve, S. (1987). 'Optimal portfolio and consumption decisions for a "small investor" on a finite horizon.' *SIAM Journal of Control and Optimization*, vol. 25, pp. 1557–86.
- King, M. and Leape, J. (1987). 'Asset accumulation, information, and the life cycle.' NBER working paper no. 2392.
- Koo, H. (1995). 'Consumption and portfolio selection with labor income I: Evaluation of human capital.' manuscript, Washington University.
- Mankiw, N. G. and Zeldes, S. (1991). 'The consumption of stockholders and non-stockholders.' *Journal of Financial Economics*, vol. 29, pp. 97–112.
- MacCurdy, T. (1982). 'The use of time series processes to model the error structure of earnings in longitudinal data analysis.' *Journal of Econometrics*, vol. 18, pp. 83–114.
- Mehra, R. and Prescott, E. (1985). 'The equity premium: a puzzle.' *Journal of Monetary Economics* vol. 15, pp. 145–61.
- Merton, R. (1969). 'Lifetime portfolio selection under uncertainty: the continuous-time case.' *Review of Economics and Statistics*, vol. 51, pp. 247–57.
- Merton, R. (1971). 'Optimum consumption and portfolio rules in a continuous-time model.' *Journal of Economic Theory*, vol. 3, pp. 373–413.
- Mossin, J. (1968). 'Optimal multiperiod portfolio policies.' *Journal of Business*, pp. 215–29.
- Polkovnichenko, V. (1998). 'Heterogeneity and proprietary income risk: implications for stock market participation and asset prices.' manuscript, Kellogg Graduate School of Management.
- Poterba, J. (1993). 'Who owns corporate stock?: a report to the New York stock exchange,' manuscript, MIT.
- Poterba, J. and Samwick, A. (1997). 'Household portfolio allocation over the life cycle.' NBER Working Paper No. 6185.
- Pratt, J. W. and Zeckhauser, R. (1987). 'Proper risk aversion.' *Econometrica*, vol. 55, pp. 143–54.
- Ryder, H. and Heal, G. (1973). 'Optimal growth with intertemporally dependent preferences.' *Review of Economic Studies*, vol. 40, pp. 1–31.
- Saito, M. (1995). 'Limited market participation and asset pricing.' manuscript, University of British Columbia, Dept. of Economics.
- Samuelson, P. A. (1963). 'Risk and uncertainty: the fallacy of the law of large numbers.' *Scientia*, vol. 98, pp. 108–13.
- Samuelson, P. A. (1969). 'Lifetime portfolio selection by dynamic stochastic programming.' *Review of Economics and Statistics*, vol. 51, pp. 239–46.

- Skiadas, C. and Schroder, M. (1997). 'Optimal consumption and portfolio selection with stochastic differential utility.' manuscript, Northwestern University.
- Sundaresan, S. (1989). 'Intertemporally dependent preferences in the theories of consumption, portfolio choice and equilibrium asset pricing.' *Review of Financial Studies*, vol. 2, pp. 73–89.
- Svensson, L. (1989). 'Portfolio choice with non-expected utility in continuous time.' *Economics Letters*, vol. 30, pp. 313–7.
- Svensson, L. and Werner, I. (1993). 'Nontraded assets in incomplete markets.' *European Economic Review*, vol. 37, pp. 1149–68.
- Tauchen, G. and Hussey, R. (1991). 'Quadrature based methods for obtaining approximate solutions to nonlinear asset pricing models.' *Econometrica*, vol. 59, pp. 371–96.
- Vila, J.-L. and Zariphopoulou, T. (1997). 'Optimal consumption and portfolio choice with borrowing constraints.' *Journal of Economic Theory*, vol. 77, pp. 402–31.
- Xu, G.-L. and Shreve, S. (1992). 'A duality method for optimal consumption and investment under short-selling prohibition II. general market coefficients.' *Annals of Applied Probability* vol. 2, pp. 87–112.
- Viciera, L. (1997). 'Optimal portfolio choice for long-horizon investors with nontradable labor income.' manuscript, Harvard University.
- Vissing-Jorgensen, A. (1997). 'Limited stock market participation.' manuscript, MIT Dept. of Economics.